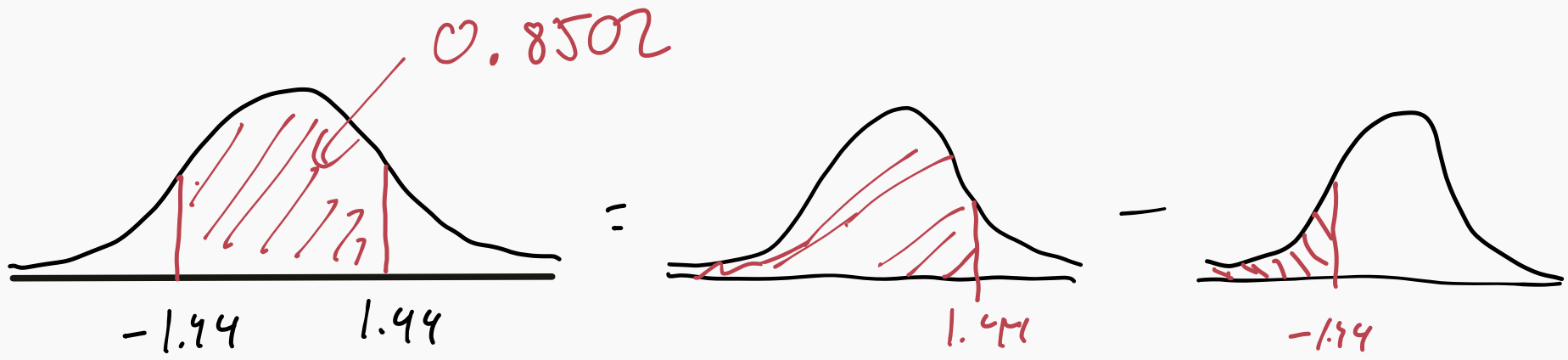
$H_0: \mu = 402$

$H_a: \mu \neq 402$

6.2



$P(-1.44 < z < 1.44) = P(z < 1.44) - P(z < -1.44)$

$= .9251 - .0749$

$= 0.8502$

8.1

$\bar{x} \mp z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$

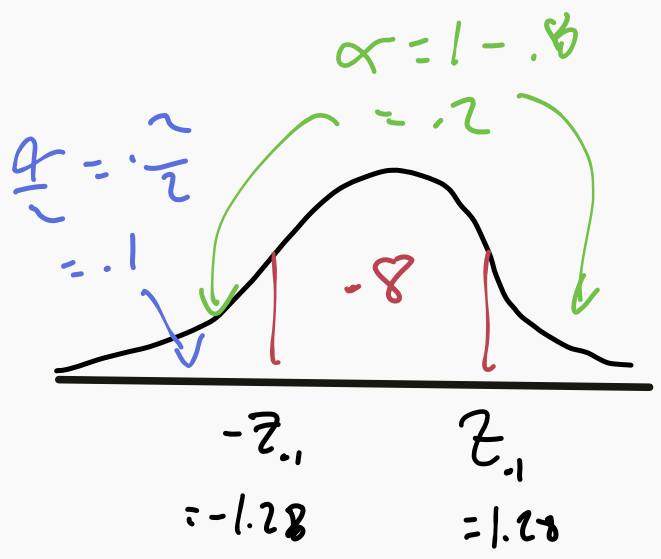
$\bar{x} = 18.1$

$\sigma = 2.3$

$n = 386$

80% CI

crit val: $z_{.1} = 1.28$



$18.1 \mp 1.28 \frac{2.3}{\sqrt{386}}$

$= 18.1 \mp 0.149846$

$(17.950, 18.250)$

$(18.0, 18.2)$

8.3

$\bar{x} \mp t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$

$\bar{x} = 57.22$

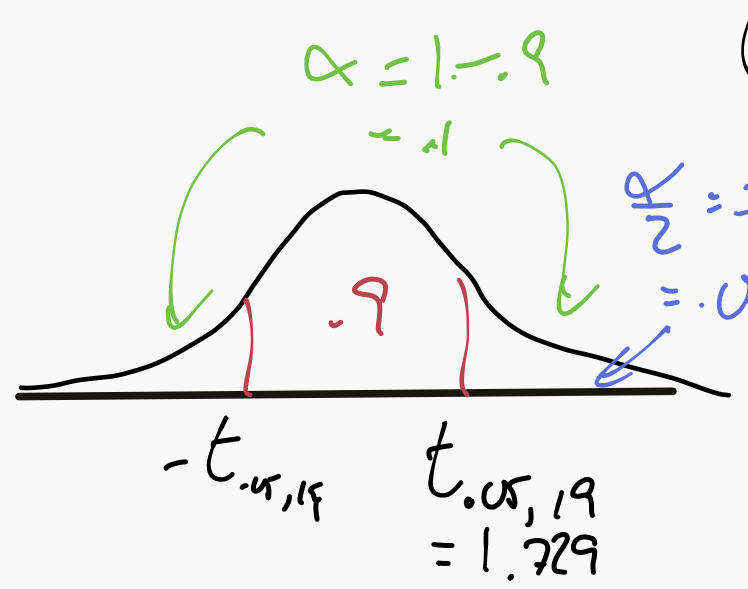
$s = 25.76$

$n = 20$

$df = 20 - 1 = 19$

90% CI

crit val: $t_{.05,19} = 1.729$



$57.22 \mp 1.729 \frac{25.76}{\sqrt{20}}$

$= 57.22 \mp 9.959232$

$= (47.2068, 67.1792)$

$(47.21, 67.18)$

7.2

$\mu = 873$

$\sigma^2 = 17161$

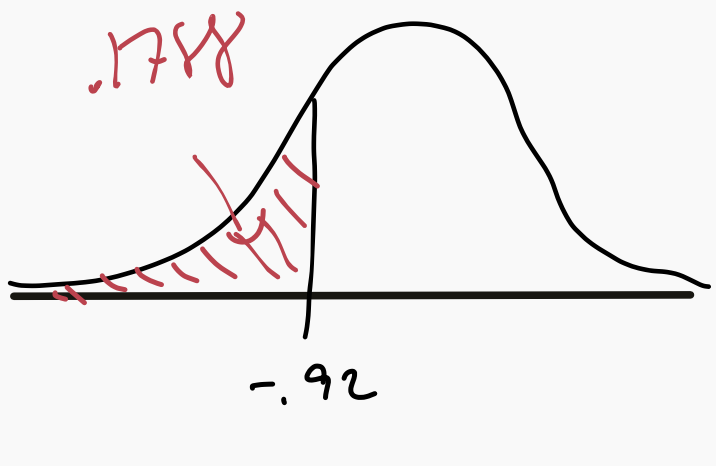
$\sigma = \sqrt{17161} = 131$

$n = 50$

$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ (7.2)

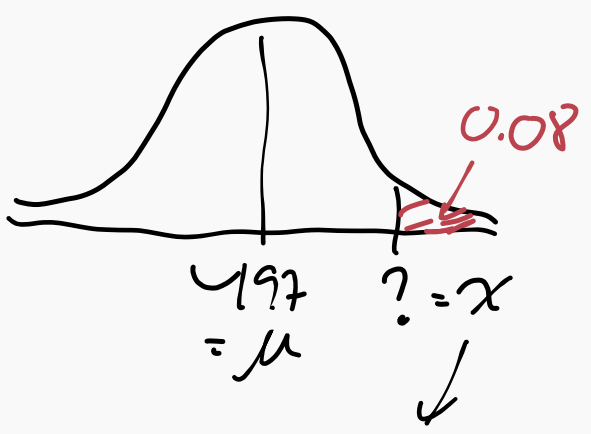
$z = \frac{x - \mu}{\sigma}$ (6.3)

$P(\bar{x} < 856) = \left[z = \frac{856 - 873}{\frac{131}{\sqrt{50}}} = -.92 \right]$



$= P(z < -.92) = 0.1788$

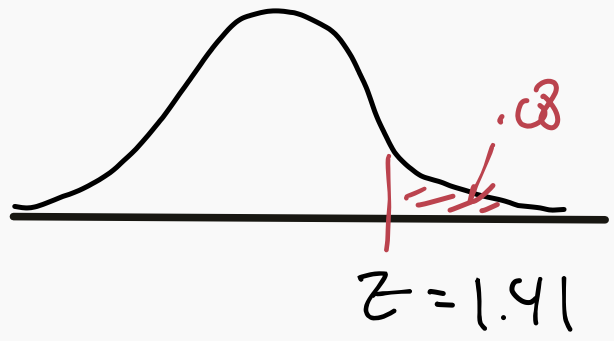
6.4



$\sigma = 109$

$z = \frac{x - \mu}{\sigma}$

$x = \mu + z\sigma$



$x = 497 + 1.41(109)$

$= 650.69$

$= 651$